



Predicting 30-Day Hospital Readmission

Using Machine Learning Techniques

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Abstract

Hospital readmissions within 30 days of discharge remain a persistent financial and clinical challenge for healthcare systems. In 2019, approximately 3.8 million patients, roughly 14 percent of all discharges, were readmitted, with each case costing an average of \$15,000–\$16,000. Beyond the monetary burden, frequent readmissions are closely linked to poorer patient outcomes.

This study develops and evaluates machine learning models to predict 30-day hospital readmission for patients with diabetes using the University of California, Irvine Diabetes 130-US dataset, which includes over 100,000 hospital encounters collected between 1999 and 2008. After extensive data cleaning and feature engineering to remove sparse or information-leaking variables and to group encounters by patient, eight models were trained, validated, and tested using a five-fold cross-validation.

Model performance was evaluated using both traditional discrimination metrics (ROC-AUC, precision, recall, F1) and a cost-based framework that links prediction thresholds to estimated per-patient savings. The expected savings function incorporated the average readmission cost ($R = \$16,300$), per-patient monitoring cost ($c_m = \$330$), and monitoring effectiveness ($e = 0.33$) to identify thresholds that maximize net benefit. Out of the model suite, XGBoost achieved the highest ROC-AUC (0.650), closely followed by RF (0.647), with KNN achieving the lowest score for this metric (0.594). The XGBoost model achieved the highest estimated per patient savings (\$37.69) by maintaining a slight edge on precision at the chosen threshold.

These results demonstrate that even modestly discriminative models can deliver meaningful economic value when evaluated through a cost-sensitive lens, emphasizing the importance of integrating financial impact into healthcare model assessment.

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Chapter 1

Introduction

1.1 Background

Hospital readmission represents a persistent and costly challenge within the healthcare system, reflecting both the quality of inpatient care and the effectiveness of post-discharge follow-up. Readmissions within 30 days of discharge are often used as a benchmark for hospital performance, patient safety, and care coordination. In October 2012, after our data’s collection period, the United States federal government created the Hospital Readmissions Reduction Program (HRRP) to incentivize hospitals to reduce their readmission rates (James, 2013). In the United States alone, unplanned readmissions account for tens of billions of dollars in avoidable healthcare spending each year, particularly among patients with chronic illnesses such as diabetes and heart disease (Jiang & Hensche, 2023). These events not only strain limited hospital resources but also signal potential deficiencies in treatment planning, discharge procedures, or access to outpatient care.

1.2 Data Science Problem

Hospital readmissions occurring within 30 days of discharge are the most suitable for prediction and prevention, as they often stem from inadequate inpatient care, poor hospital-to-home transition support, or insufficient post-discharge follow-up (Benbassat & Taragin, 2000).

Therefore, the goal of our project is to predict, at the time of discharge, whether a given patient will be readmitted to the hospital within 30 days. Patients flagged by a machine learning model as “high risk” could be placed on a monitoring program or receive additional post-discharge care from the hospital to reduce the likelihood of a near-term readmission.

Formally, we frame this as a binary classification task that predicts whether an encounter will result in a 30-day readmission (`readmitted` = <30) versus not (`readmitted` = >30 or NO).

1.3 Project Summary

In this study, we use the UCI Diabetes 130-US dataset to train and evaluate seven traditional machine learning models, and one neural-network, to predict 30-day readmission among diabetic patients. We perform extensive preprocessing and feature engineering based on prior work, apply group-aware cross-validation to avoid patient-level leakage, and compare models using both ROC-AUC and a cost-based decision framework that estimates per-patient savings under a remote patient monitoring program.

1.4 Previous Literature

Several studies have previously been conducted in the area of attempting to predict patients who will be readmitted over short- and long-term durations, with limited success. Access to suitable large-scale datasets that involve patient health records remains a difficulty, although this is being addressed with more modern tools, such as the widespread adoption of Electronic Health Records (EHR).

Previous studies, such as those performed by Emi-Johnson and Nkrumah (2025) and Liu et al. (2024), list results for several traditional machine learning models, including Random Forest (RF), Support Vector Machines (SVM), Extreme Gradient Boosting (XGBoost), and Logistic Regression. These studies report an area under the receiver operating characteristic curve (ROC-AUC)—a key measure of class discrimination—ranging between 0.57 and 0.667, depending on the model. This provides an excellent starting point for model selection and serves as a benchmark for evaluating our own performance.

Numerous studies have also examined the effectiveness of remote patient monitoring (RPM) in reducing hospital readmissions. Estimates from this research serve as the foundation for our analysis and can be further refined through future clinical validation.

Chapter 2

Methods

2.1 Dataset Description

The dataset used in this study is the University of California, Irvine (UCI) Diabetes 130-US dataset. This dataset represents ten years (1999–2008) of patient data across 130 hospitals in the United States. It was originally compiled for the study by Strack et al. (2014). Each row of the dataset corresponds to a single inpatient encounter with a patient diagnosed with diabetes, during which the patient underwent at least one laboratory test and received medication while hospitalized.

The data consists of 50 features. Some identifiers, such as patient number and encounter ID, are included, but most features pertain to patient and hospital outcomes. These include patient demographics such as race, gender, and age, as well as treatment-related information such as length of stay, type of admission and discharge, and medication changes during hospitalization.

The class variable was derived from the `readmitted` column in the dataset, which originally contained three values: `<30`, `>30`, and `NO`, indicating readmission within 30 days, after 30 days, and no readmission, respectively. The column was transformed into a binary integer variable, equal to 1 when `readmitted` = `<30`, and 0 otherwise.

2.2 Data Quality

The dataset exhibits several quality issues that must be addressed during preprocessing. Some columns are extremely sparse, such as `weight`, which is missing approximately 97% of its values. Other columns contain categorical variables with very high cardinality, exceeding 900 distinct categories. Finally, several features leak information that would not be available at the time of discharge, such as `discharge_disposition`, which includes cases where patients were known to have left the hospital and subsequently passed away.

2.3 Exploratory Data Analysis

Exploratory data analysis confirmed the quality issues described above, enabled detailed inspection of categorical variables, and supported statistical assessment of continuous features for potential predictive signal.

Analysis of the `patient_nbr` column revealed that several rows represented repeated visits by the same patient. The dataset was also examined for missing values. The `weight` column was found to contain 96.86% missing values, while other columns such as `max_glu_serum`, `A1Cresult`, `medical_specialty`, and `payer_code` also exhibited nearly 50% or more missing data.

Statistical analysis showed that several continuous features were heavily right-skewed, particularly those representing counts of inpatient hospital visits, which contained a large proportion of zeros. Further inspection of continuous variables did not reveal any major data integrity issues, such as negative values in features that should only contain positive numbers.

2.4 Data Cleaning

The following features were dropped: `encounter_id` (identifier) and `payer_code` (non-predictive). The `patient_nbr` feature is used for grouping and then dropped before model training.

The target population for this study is adult diabetic patients who were discharged from the hospital. Therefore, the rows corresponding to newborns, deceased patients, and patients transferred to another inpatient facility were removed. After this data cleaning process, 78,521 encounters remained.

2.5 Feature Transformation

We used the binning strategy employed by Strack et al. (2014) in the original study accompanying the dataset. That is, to combine several of the categories of many of the features to reduce complexity and noise. Our binning strategy diverged in several areas, as described below. The full binning strategy can be found in Appendix A.

The A1C test results were subdivided into three categories, unlike the study by Strack et al. (2014), whose main focus was on the relationship between the A1C test results and readmission and, therefore, a slightly different focus. Glucose test results were binned in the same way as A1C results (*high*, *normal*, *no-test*), with the majority of both categories being *no-test*.

This strategy of binning also mitigates a small number of missing values in the `race` (2.2%), `diag_3` (1.4%), and `diag_2` (0.3%) columns by imputing with default values of *other* in each case.

Once the binning strategy was completed, the new categories are one-hot or ordinal encoded accordingly.

Medication columns `change` and `diabetesMed` were simplified to a binary zero-or-one indicator. The 23 medication column features were also simplified to a binary flag and summed across them to create a `num_drugs` column.

Several columns were then dropped from the dataset: the medications and the medication flags `insulin` and `metformin`, as these are the two most common medications used for long-term diabetes treatment (U.S. Dept. HHS, 2022). The most common column from the one-hot encoded categoricals was also dropped to prevent perfect collinearity between columns.

Among the continuous features, the three features relating to number of hospital visits have heavily right-skewed histograms and contain a vast majority of zero values.

These features `number_inpatient`, `number_outpatient` and `number_emergency`, were binned into five categories: *zero*, *one*, *few*, *several*, and *frequent* and then ordinal encoded.

Finally, for linear models (SGD and SVC), a log transformation was applied to all continuous features except `num_lab_procedures`, as that feature already exhibited a bell-shaped distribution during exploratory data analysis.

2.6 Feature Engineering

The dataset is provided as-is, after the previously mentioned transformations, to the tree models, but several interaction terms were created for the linear models.

Several interaction terms were found to be statistically significant in the study by Strack et al. (2014). The interactions in table 2.1 were chosen to begin with, as they display the highest statistical significance.

Table 2.1: Interaction Terms Selected Based on Statistical Significance

Variable 1	Variable 2	p-value
Discharge disposition	Time in hospital	< 0.001
Medical specialty of admitting physician	Age	< 0.001
Primary diagnosis	Time in hospital	< 0.001
Discharge disposition	Race	< 0.001
Discharge disposition	Medical specialty of admitting physician	0.001
Medical specialty	Time in hospital	0.001

2.7 Analysis Strategy

The first model selected was the Random Forest, chosen for its high ROC-AUC performance reported in studies by Liu et al. (2024) and Mahmoud et al. (2023). It also offered an interpretable means to assess the importance of features early in the process to evaluate the effectiveness of feature engineering.

All models chosen are listed below in Table 2.2

Each model was trained and evaluated using the same procedure. The dataset was split into training (70%), validation (15%), and testing (15%) subsets, stratified by class and grouped by `patient_nbr`. A baseline model was computed using a zero-parameter

Table 2.2: Model Selection

Model	Reasoning
Random Forest	RF showed very good performance in previous studies and provides a natural way of viewing feature importance.
Stochastic Gradient Descent	scales efficiently to high-dimensional data and provides a fast, linear baseline that incorporates regularization
Support Vector Machine	SVM was included because it provides a strong, theoretically grounded non-linear classifier that often performs well on medium-sized tabular datasets.
Extreme Gradient Boosting	XGBoost was chosen after previous studies had found it to be the best performing model of their tested sets.
K-Nearest Neighbors	KNN was included as a baseline non-parametric model to contrast with linear methods and for its simplicity
Logistic Regression	provides a well-understood probabilistic framework for binary classification, making it an ideal benchmark for evaluating more complex models
Multilayer Perceptron	naively chosen for its ability to model complex non-linear relationships and usage in previous studies

configuration with `cross_val_score`, employing ROC-AUC as the metric and a five-fold group-aware cross-validation (CV). This baseline also enabled testing of oversampling versus undersampling techniques to address class imbalance.

Subsequently, each model underwent randomized hyperparameter tuning, also using five-fold group-aware CV, to determine optimal parameter configurations. A complete list of tuned hyperparameters for each model is provided in Appendix B. The tuned models were then validated and tested.

Finally, the models were used to generate predictions on the held-out validation set. These metrics were used to compute a cost-weighted decision threshold, after which each model generated predictions on the held-out test set. Metrics were recorded at each stage for baseline, validation, and test results, and optimal hyperparameters.

2.8 Cost-Based Evaluation Framework

To evaluate our models within a business context, we introduce a cost-weighted threshold designed to provide the optimal economic outcome for the hospital. We define a hypothetical Remote Patient Monitoring (RPM) program to which patients flagged by the model would be enrolled.

Estimating cost savings per patient can be done through the following process. We define several parameters, either known or estimated:

- R : the average cost of a hospital readmission,
- c_m : the monitoring cost per flagged patient,

- e : the effectiveness of the monitoring program at preventing readmissions,
- r : the percent of patients whose readmissions are preventable,
- p : the baseline readmission rate, and
- model metrics at a given prediction threshold t : recall (or $TPR(t)$), precision ($PPV(t)$), and false positive rate ($FPR(t)$).

The expected savings per flagged patient at threshold t is given by:

$$S_{pp}(t) = PPV(t) \cdot r \cdot e \cdot R - c_m$$

where $S_{pp}(t) > 0$ indicates that each flagged patient is, on average, profitable at that threshold.

The total expected savings across the patient population is defined as:

$$S(t) = p \cdot TPR(t) \cdot reR - c_m \cdot [p \cdot TPR(t) + (1 - p) \cdot FPR(t)]$$

where $S(t)$ is maximized by selecting the optimal threshold t .

The term in brackets,

$$p \cdot TPR(t) + (1 - p) \cdot FPR(t),$$

represents the *flag rate* of the model (the proportion of patients who will be monitored).

Therefore, this can be expressed as

$$S(t) = \text{flag}(t) \cdot (PPV(t) \cdot reR - c_m)$$

showing that the sign and magnitude of the net savings depend directly on input costs and model precision.

By varying the values of R , e , and c_m , we effectively create three pseudo-hyperparameters that strongly influence the model's predictions and, consequently, its accuracy, precision, recall, and F_1 scores.

The values of R (\$16,300), r (0.50), and c_m (\$330) were taken from prior research by Zhang et al. (2025). The effectiveness value e was estimated between 50–60% based on previous studies by Godla (2025), Leo et al. (2022), Po et al. (2024), Taylor et al. (2021), and Thomas et al. (2021). That value was reduced to account for potential lack of correlation between study populations and the increased difficulty of predicting readmissions across multiple conditions.

2.9 Analysis Code

[Collab Notebook Link](#)

Chapter 3

Results

3.1 Model Performance

All models exhibited similar performance. As a measurement of class distinction the ROC-AUC score for each model was recorded on the validation and test sets.

The table below shows the metric scores for each model, representing an average between results from validation and test sets.

Table 3.1: Model Performance (mean of test and validate sets)

Model	ROC-AUC
XGB	0.6566
RF	0.6511
SVC	0.6481
LR	0.6477
SGD	0.6477
MLP	0.6048
KNN	0.5839

3.2 Cost-Based Performance

An estimated per-patient savings was calculated for the test set using the validation-calculated threshold. When holding R , e , and c_m equal, we can compare the models against each other to maximize savings. The table below shows estimated per-patient savings for each model given $R = \$16,300$, $e = 0.33$, and $c_m = \$330$. Metrics from the held-out test set are also reported.

Given our input variables, we can calculate a break-even precision of 6%. Each model easily achieves this metric, with SGD slightly pulling ahead in per-patient savings due to a smaller amount of false positive predictions.

Table 3.2: Est. Per-Patient Savings and Performance Metrics (Test Set)

Model	Est. PP Savings (\$)	ROC-AUC	Recall	Precision
XGB	37.69	0.650	0.346	0.190
LR	37.27	0.640	0.349	0.191
SVC	35.82	0.639	0.384	0.176
SGD	35.39	0.640	0.363	0.183
RF	35.02	0.647	0.406	0.175
MLP	20.38	0.608	0.293	0.154
KNN	8.40	0.594	0.235	0.150

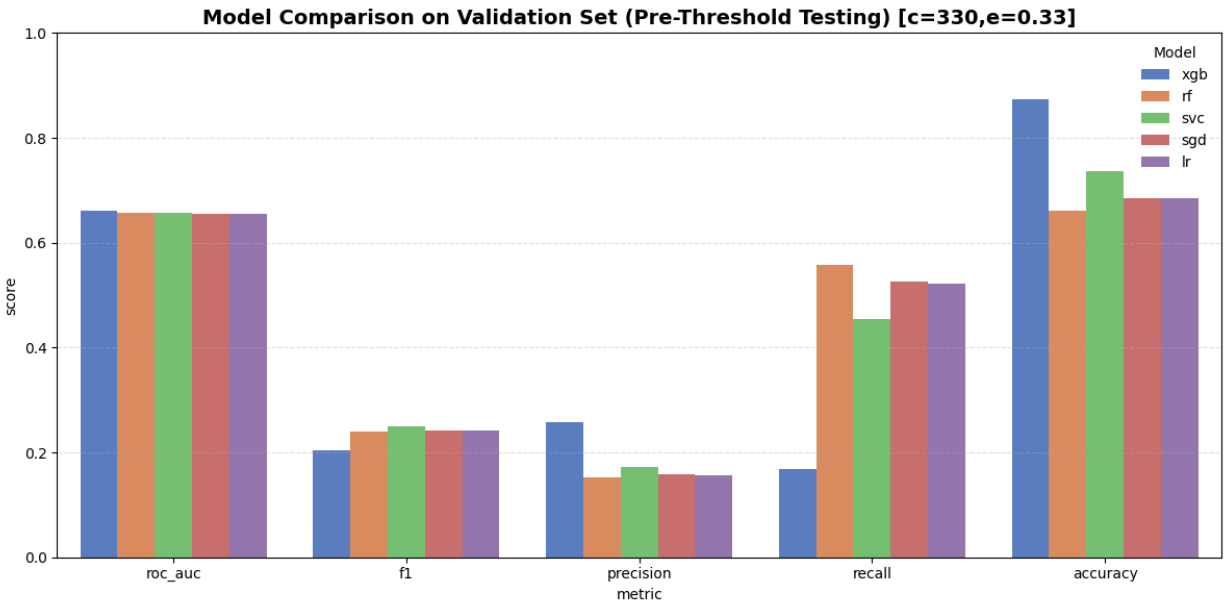


Figure 3.1: Held-out Validation Set Metrics

3.3 Feature Importance

Permutation importance of features was run for each model, except SVC, due to computation time. Tree-based feature importance was ruled out early on, as it heavily favors continuous features over categorical ones. The tree-based feature importance results can be seen in Appendix D.

Across all three models, the variable `number_inpatient` consistently dominated as the single most influential predictor of 30-day hospital readmission. Beyond this leading variable, there is general agreement across models regarding the next tier of influential features. `number_diagnoses`, `num_medications`, `num_lab_procedures`, and `time_in_hospital` frequently appear near the top of the rankings.

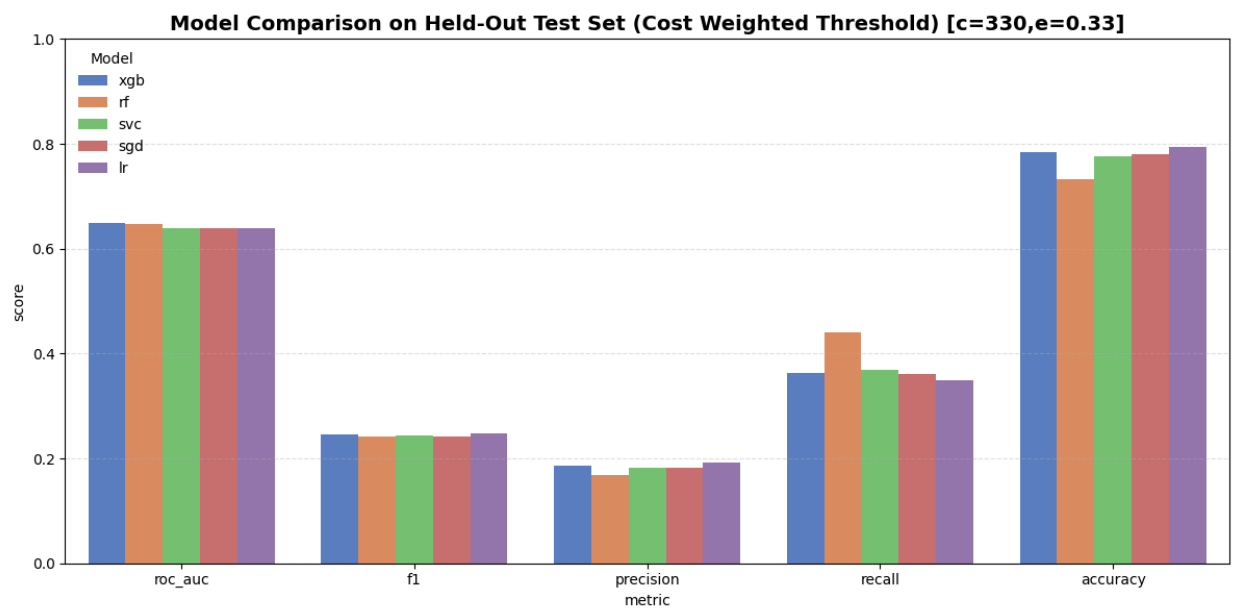


Figure 3.2: Held-out Test Set Metrics

Chapter 4

Discussion

4.1 Model Performance and Feature Importance

Across models, performance metrics exhibited consistent behavior, with only marginal variation in ROC-AUC values but noticeable shifts in downstream metrics once the cost-based decision threshold was applied. ROC-AUC values ranged narrowly between 0.647 and 0.650, indicating that all models captured a similar level of separability between readmitted and non-readmitted patients.

From an economic perspective, the XGBoost model achieved the best overall performance by achieving the highest precision at the chosen cost threshold.

MLP and KNN fell behind the other models in terms of ROC-AUC and therefore achieved lower precision, recall, and estimated savings. The MLP's under-performance is consistent with literature regarding sparse, tabular, one-hot encoded features (Shwartz-Ziv & Armon, 2021) and KNN has shown to perform poorly on high-dimensional sparse data (Kononenko, 2001) showing both models are poor choices for this classification task.

In terms of feature importance, the leading feature, `number_inpatient`, which measures the number of prior inpatient visits, emerged as the most influential variable. Its strong predictive power aligns with clinical expectations: frequent hospitalizations often indicate chronic or poorly managed conditions and are a key risk factor for early readmission.

Taken together, the feature importance results suggest that prior healthcare utilization and overall illness complexity are the principal determinants of readmission within this dataset, whereas demographic or diagnostic subgroup variables play secondary roles. The consistency of these findings across models reinforces their robustness.

The metrics heatmap (Figure 4.1) illustrates that when the program monitoring cost (c_m) increases or program effectiveness (e) decreases, the break-even precision threshold rises, prompting the model toward stricter classification behavior. In these scenarios, XGB slightly outperforms the other models by achieving the highest precision. These observations underscore the importance of accurately determining c_m and e for the specific monitoring program.

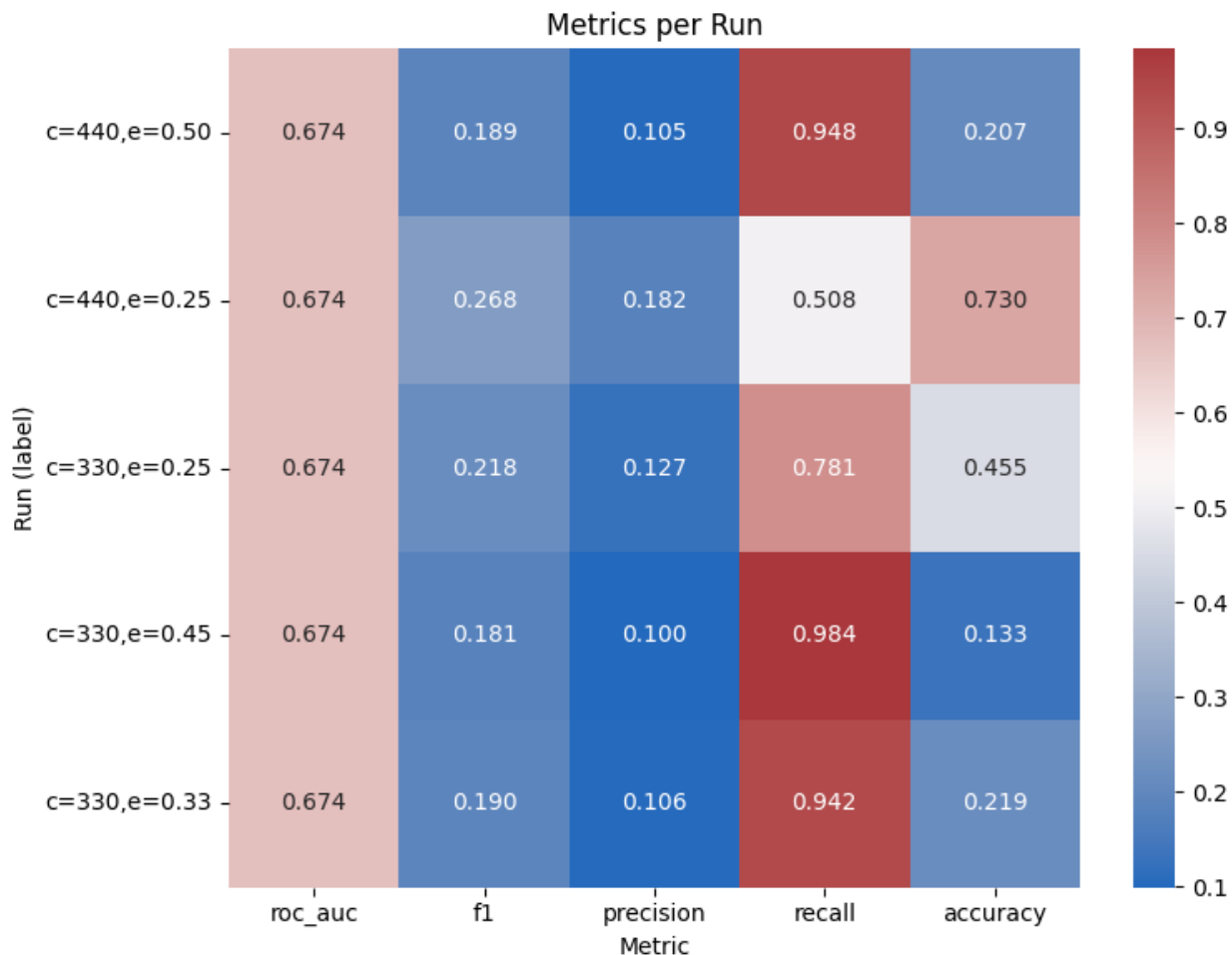


Figure 4.1: Inter-run metrics heatmap

4.2 Relation to Prior Work

Feature importance results align closely with findings from previous research and clinical expectations. Studies such as Emi-Johnson and Nkrumah (2025) have also identified `time_in_hospital`, `number_inpatient`, and specific diabetic medications, including insulin prescriptions, as key predictors of hospital readmission.

Model performance metrics similarly correspond with prior literature. Other studies, including Emi-Johnson and Nkrumah (2025) and Mahmoud et al. (2023), report comparable ROC-AUC values when evaluating models on the same dataset, strengthening the validity of our results.

4.3 Limitations

This study relies on data from 1999–2008, which may not fully reflect current clinical practices or readmission patterns. The UCI Diabetes 130-US dataset contains many sparse

features and lacks the depth of information available in modern electronic health records (EHRs), such as detailed laboratory results and vital signs, thereby limiting predictive performance. Additionally, the dataset exhibits a highly imbalanced class distribution, with roughly one 30-day readmission for every ten patient encounters. This imbalance complicates model training and evaluation. The observed readmission rate is also approximately 33% lower than rates reported in more recent studies on the same topic.

Because the cost-based threshold approach depends heavily on the assumed parameters—readmission cost (R), preventable readmission percentage (r), program effectiveness (e), and monitoring cost per patient (c_m)—the estimated savings should be interpreted as illustrative rather than definitive. These values would require site-specific validation for real-world applicability.

Using the cited parameters for monthly monitoring cost (\$330), assumed program effectiveness (33%), and average readmission cost (\$16,300), the breakeven precision threshold is only 12%, a level cleared by all models, although not by much. While the models maximize estimated savings under these assumptions, the relatively low monitoring cost and high program effectiveness compared to readmission cost suggest that monitoring all patients in this dataset would likely be cost-effective. Although this approach would lower per-patient savings, it would likely produce the greatest overall improvement in patient outcomes.

The findings suggest that machine learning models for predicting hospital readmissions can be most useful when viewed through a financial lens. In practice, hospitals could adjust the cost parameters based on their own institutional data, to create a personalized model that could maximize that facility’s savings.

Chapter 5

Conclusion and Future Work

This project developed and evaluated several machine learning models for predicting 30-day hospital readmission among diabetic patients, incorporating a cost-sensitive framework to link model outputs to potential financial savings. While all models achieved comparable discrimination, results showed that optimizing for cost efficiency rather than raw predictive power can meaningfully alter which model performs “best.” In this case, the XGBoost classifier delivered the highest estimated per-patient savings by maximizing precision at the chosen threshold, underscoring the need to evaluate healthcare models through an operational and economic lens rather than a purely statistical one.

Future work should focus on validating these findings with more recent and comprehensive patient data, integrating additional clinical features such as lab values, vitals, and adherence to medications. Testing site-specific parameter values for readmission cost, monitoring expense, and intervention effectiveness would provide more realistic savings estimates.

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Appendix A

Full Binning Strategy

Table A.1: Feature Binning Summary

Feature	Type	Original Count	Binned Count
diag_1, diag_2, diag_3	Nominal	>900	9
admission_source_id	Nominal	17	3
discharge_disposition_id	Nominal	26	4
admission_type_id	Nominal	8	5
medical_specialty	Nominal	73	6
race	Nominal	6	3
age	Ordinal	10	3
A1C results	Ordinal	4	3
max_glucose_serum	Ordinal	4	3
number_emergency	Continuous → Ordinal	22	5
number_outpatient	Continuous → Ordinal	37	5
number_inpatient	Continuous → Ordinal	21	5

Appendix B

Hyperparameter Tables

Table B.1: XGBoost Hyperparameters and Interpretation

Parameter	Value	Explanation	Interpretation of Value
subsample	0.6	% of training data sampled per tree	Low value, may be due to correlated features. Fights overfitting.
scale_pos_weight	4.0	Scale of positive class weight, replaces resampling methods for this model	About 50% of the actual class imbalance.
min_child_weight	2.0	Minimum sum of instance weights in a leaf	The model found useful patterns in smaller subsets.
learning_rate	0.1	Step size	Model converged more slowly.
gamma	0	Minimum loss reduction required to make a split	Split freely.
colsample_bytree	0.8	% of features used per tree	80% of features used per tree.

Table B.2: Support Vector Classifier Hyperparameters and Interpretation

Parameter	Value	Explanation	Interpretation of Value
C	2.397	Inverse regularization strength	Attempts to fit data tightly.
gamma	0.001	Kernel curvature strength	Broad, smooth decision boundary.
kernel	rbf	Nonlinear kernel	Common kernel for non-linear separation.

Table B.3: Stochastic Gradient Descent Hyperparameters and Interpretation

Parameter	Value	Explanation	Interpretation of Value
tol	0.0001	When to stop learning	Precise training.
shuffle	True	Randomize training samples each epoch	Standard.
penalty	l1	Lasso regularization	Helps reduce less useful features.
max_iter	500	Max number of data passes	Needs some work to converge.
loss	modified_huber	Hybrid hinge + squared loss	Robust to outliers, ideal for noisy clinical datasets.
learning_rate	invscaling	Shrink learning rate over time	Speeds up stability.
l1_ratio	0.9	Increase penalty	Increases feature pruning.
eta0	0.003	Initial learning rate	Keeps early gradients stable.
class_weight	None	If the classes need to be balanced after resampling	None makes sense after RandomUnderSampler.
average	10	Uses an average of coefficients for stability	Helps reduce variance with noisy data/gradients.
alpha	0.001	Regularization strength	Less penalty, helps model generalize after heavy feature pruning from L1.

Appendix C

Permutation Importance Figures

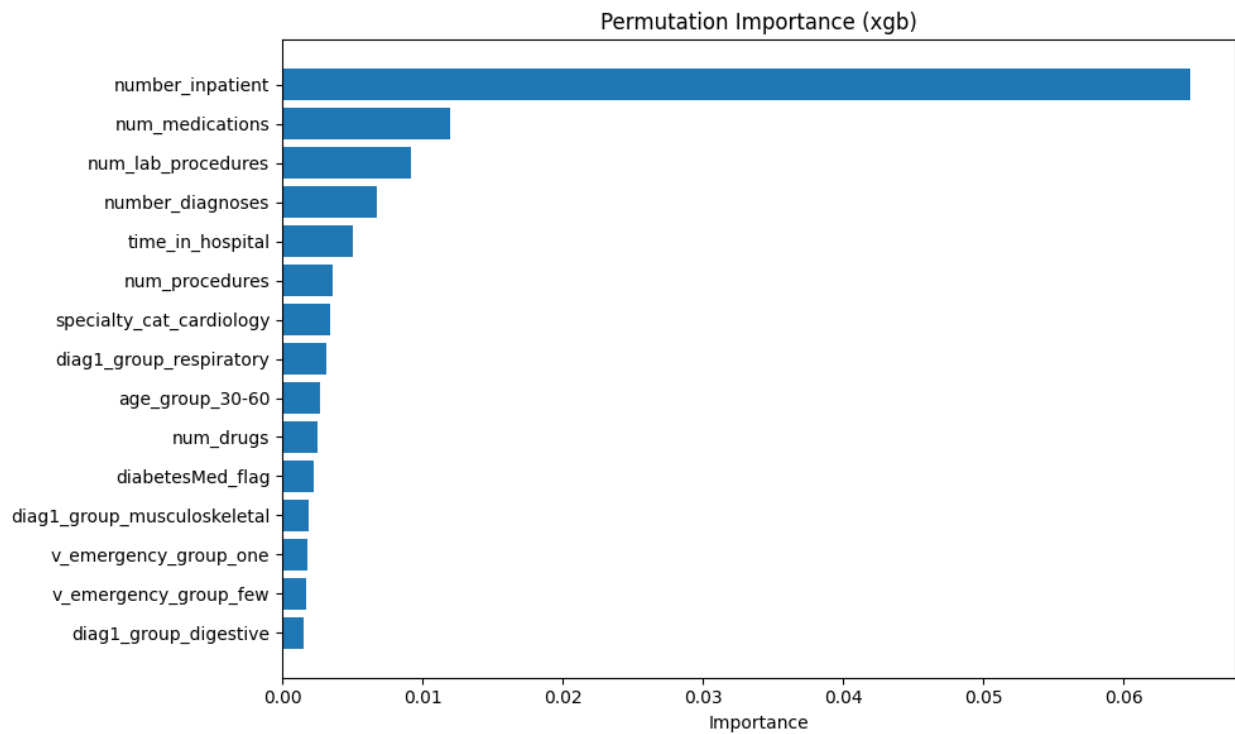


Figure C.1: XGB Feature Importance

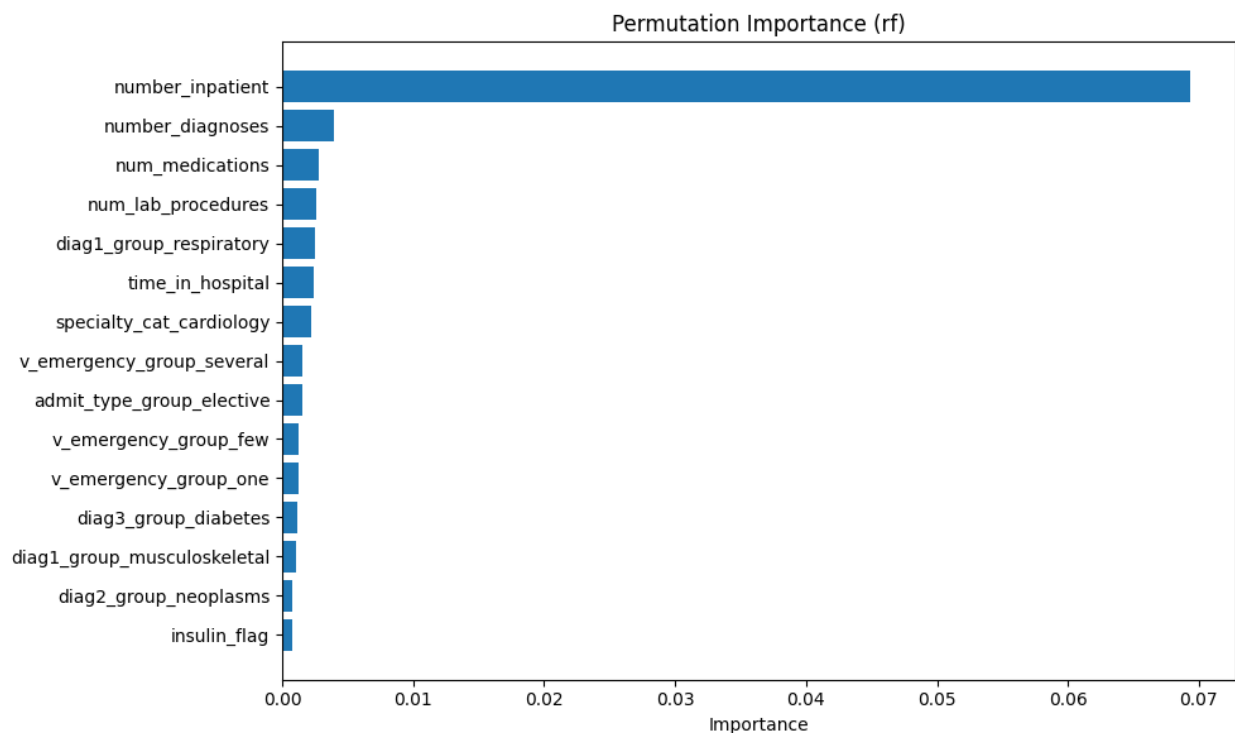


Figure C.2: Random Forest Feature Importance

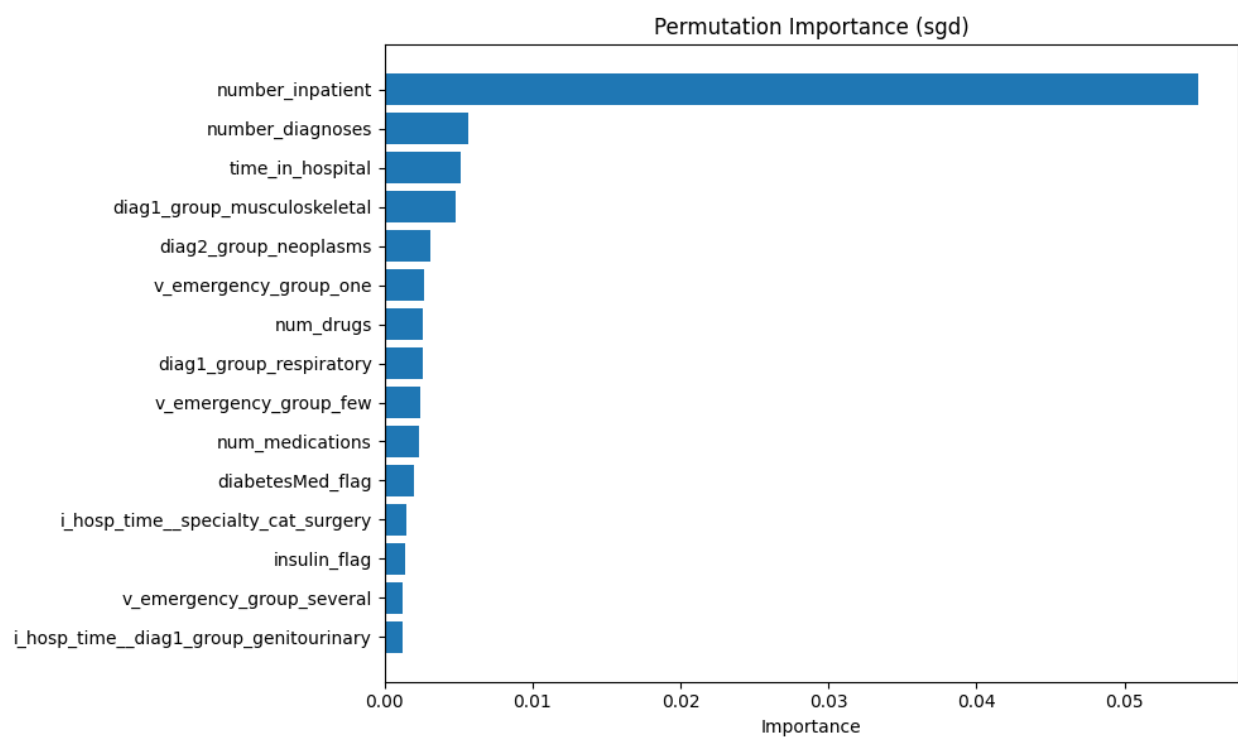


Figure C.3: SGD Feature Importance

Appendix D

Tree-Based Feature Importance

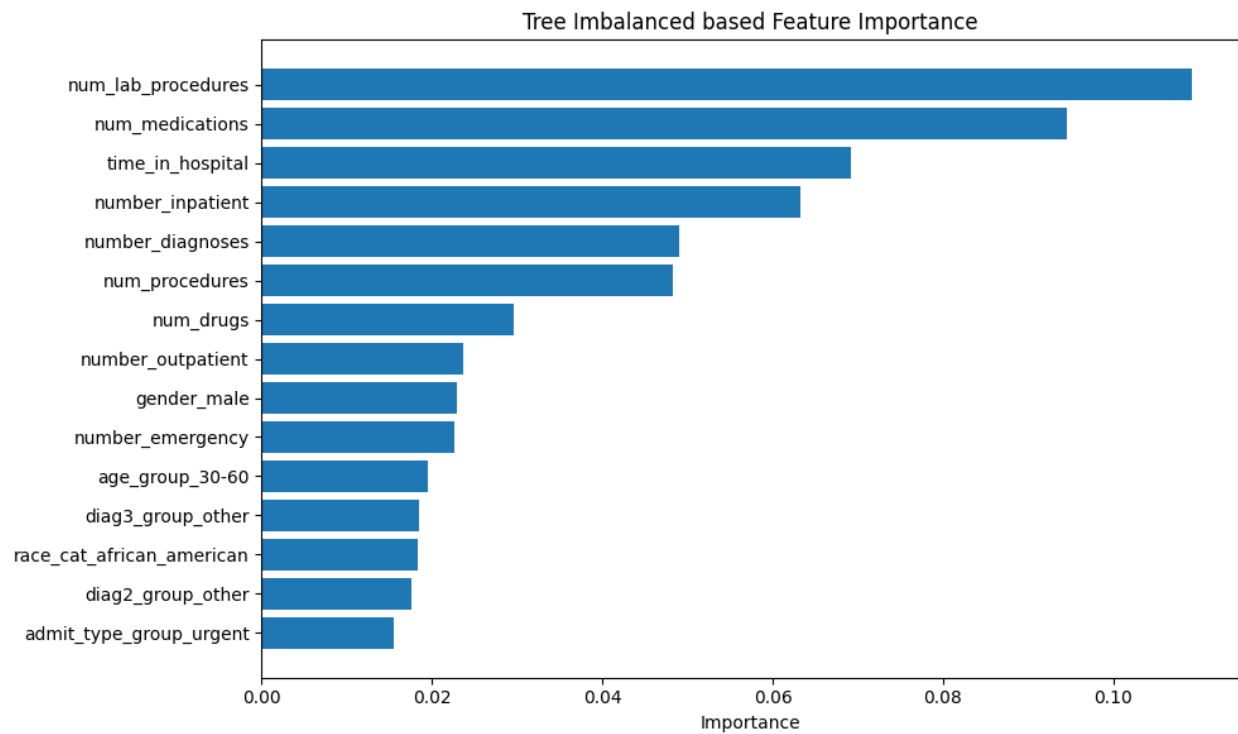


Figure D.1: Tree Based Feature Importance